How to Approach This Project

Step-by-step guide

- ➔ **Download the dataset and explore it thoroughly.**
  Since you already know both SQL and Python, this should be quite manageable for you
- ➔ **Do some research before jumping in.**
  Look at how similar projects are built examples like Zomato, Amazon, etc.
  You can explore Google, YouTube, blogs, or even your favorite AI tools to understand real-world use cases and project structures.
- ➔ **Understand the dataset deeply.**
  Go through each column carefully and understand why it exists and what it represents.
  Based on this understanding, clean the dataset wherever required (missing values, duplicates, incorrect data types, etc.).
- ➔ **Build relationships between tables.**
  Once the data is clean, review the questions provided in the file and identify how different tables relate to each other.
  Creating an **ERD (Entity Relationship Diagram)** is highly recommended, it will give you much better clarity.
- ➔ **Start solving the questions using SQL.**
  Take your time to think through each question and how it connects to the dataset.
  Don't rush to AI, try to solve it on your own first. This thinking process is where real learning happens.
- ➔ **Do some additional analysis on your own.**
  After completing all the given questions, explore the data further and perform extra analysis.
  This will boost your confidence and make your project stand out.
- ➔ **Create a strong and engaging presentation.**
  Explain:
  - ◆ What the project is about
  - ◆ How you approached it
  - ◆ Challenges you faced
  - ◆ Key insights you discovered
    Keep it informative, honest, and interesting.

➔ **Upload everything to GitHub.**

Include:

- ◆ The dataset
- ◆ SQL queries / Python code
- ◆ Final solutions
- ◆ A well-written and clean `README.md`

  This is very important.

➔ **Prepare for the final presentation.**

Practice explaining the project with your teammates or classmates so you're confident on presentation day.

➔ **Final and most important rule:**

Marks will be given ONLY if you present the project in the final class.

There is **no Google Form submission** just to get marks, your presentation is mandatory.

Important note for group projects

If you're working in a team:

- You should conduct Google Meet sessions to discuss roles and responsibilities.
- Every team member **must understand the entire project**, not just their own part.
- On presentation day, **any member can be asked questions** about any part of the project.
- Lack of understanding will result in **mark deduction**.

Questions

1. What are the top 10 restaurants by total sales amount?
2. What is the average rating and total rating count for restaurants in the top 20 cities?
3. What are the monthly order trends based on order volume over time?
4. What are the top 5 most popular cuisines by order volume?
5. What is the distribution of vegetarian vs non-vegetarian items ordered?
6. What are the top 20 cities by the number of restaurants?
7. How do different user demographics correlate with average order value?
8. Who are the top 15 highest-spending users?
9. What are the top 15 cuisines with the highest average menu prices?
10. Which restaurants offer the most diverse menu, based on the number of unique cuisines and dishes available?
11. What are the most ordered food items across all restaurants?
12. How does spending behavior differ between genders?
13. On which days of the week do restaurants experience peak order volumes?
14. How does order frequency vary across different income groups?

Note: These are **mandatory questions** and must be solved using **SQL**. However, you are encouraged to think beyond these questions and derive **at least 2 - 3 additional insights** based on your own observations of the dataset

Feel free to reach out on Slack if you need any help. However, I strongly encourage all of you to collaborate and work with a team spirit learning together will make this experience more powerful and impactful.